

A Data-Driven Reduced-Order Model for Friction-Stir Welding

POD/SVD Feasibility: Density Fields and Particle Clouds

FSW-ROM study (JAXTrace)

July 3, 2026

Abstract

We assess whether a reduced-order model (ROM) can predict the full 3D outcome of a friction-stir-welding (FSW) simulation from two process inputs — advance velocity v_{adv} (`INLET_VELOCITY`) and pin rotation ω_{pin} (`PIN_RPM`) — using linear Proper Orthogonal Decomposition (POD) via the SVD. Two output targets are studied: the Eulerian mean-density field and the Lagrangian particle cloud. The manifold of responses is *curved* (intrinsic dimension ~ 2.5 below ~ 10 linear modes for 90% energy), so linear POD is a *convenient first approximation, not a fitted model* of the geometry. With only 20 cases the density surrogate reaches $\sim 28\%$ held-out error (per-case spread 18–43%), and this error has *two* non-negligible parts: a ~ 10 –14% linear-POD *basis* error (the singular spectrum has no elbow — the curved ensemble is not low-rank in a linear basis) plus the additional *regression* error, the latter most simply explained by degrees-of-freedom starvation (fitting ~ 7 –10 mode coefficients from 20 points). Two representation choices reduce the error — the *time-averaged* density beats a single final snapshot, and a *2D* (y, z) *cross-section* reaches $\sim 21\%$ (with the caveat that it also predicts a lower-information target). Curvature and active-subspace diagnostics are *suggestive but not conclusive at $n=20$* : they hint that the error concentrates on the sampling boundary (extrapolation) rather than in high-curvature interior regions, but the correlations rest on ~ 20 noisy LOOCV estimates and cannot separate this from “too few samples, period.” The unambiguous, method-independent conclusion is that **more samples are required** — particularly to extend and densify the sampling envelope and to resolve where the curvature changes — both to lower the linear-POD error and to *unlock* non-linear alternatives (patch/local POD, manifold learning, autoencoders) that 20 points cannot currently support.

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1 Dataset

Twenty full-order FSW simulations on a Latin-hypercube-like sampling of $(v_{\text{adv}}, \omega_{\text{pin}})$, with $v_{\text{adv}} \in [5.0, 10.0] \times 10^{-3}$ m/s and $\omega_{\text{pin}} \in [-800, -400]$ RPM. Authoritative parameters come from each case’s `run_jaxtrace.sh` (the CSV `dt` column is inconsistent and was not used). A notable design property: $v_{\text{adv}} \cdot t_{\text{max}} = v_{\text{adv}} \cdot \text{DT} \cdot N_{\text{STEPS}} = 0.075$ for *every* case — the runs are length-matched, so all cases advect the same total distance.

	density study	particle study
snapshot	one static 3D <code>mean_density</code> field	final-step displacement of 360k particles
cases used	20	20
source	<code>particles_union_density.vtkhdf</code>	<code>particles.vtkhdf</code>

Table 1: The two ROM targets. Both use all 20 cases after data-quality fixes (Sec. 5).

2 Method: pre-SVD preparation, SVD, and validation

Correspondence (pre-SVD). POD needs every snapshot as a vector on *one shared grid*, so that “column j ” means the same thing in every case. The native density grids differ per case (extent/spacing scale with v_{adv}), so every case is trilinearly resampled onto a common intersection-box grid, then a boundary margin is trimmed (1 voxel per wall, 6 off z_{max} to remove an artificial top-layer). For particles, the seeds are byte-identical across cases (same 360k particles, same `ParticleID` order), so correspondence is free: a per-particle displacement stacks directly.

SVD (method of snapshots). The snapshot matrix $X \in \mathbb{R}^{n \times m}$ is short and fat ($n = 20$ cases, $m \sim 10^6$ voxels). We mean-centre and factor the small Gram matrix $G = X_c X_c^\top \in \mathbb{R}^{n \times n}$ with `jax.numpy.linalg.svd`, recovering the spatial modes; this is identical to the economy SVD of X_c but avoids GPU `gesvd` failures on the wide matrix. The per-voxel mean is the POD reference field (“mode 0”); optional normalizations (global scalar, per-case L^2/mass , log) were tested but change little (Sec. 3).

Validation: projection vs. LOOCV. Two error numbers appear throughout and must not be conflated. *Projection error* fits POD on all cases and reconstructs case i from its *own* coefficients — it isolates the *basis* error (what linear POD cannot represent even with perfect coefficients). For density it is $\sim 14\%$ at $K=7$ and still $\sim 10\%$ at $K=10$ (90% energy) — *not* negligible. *Leave-one-out cross-validation* (LOOCV) is the real generalization test: for each case i , fit POD on the other

$n-1$, fit a regressor $(v_{\text{adv}}, \omega_{\text{pin}}) \mapsto$ mode coefficients on them, predict the held-out coefficients, reconstruct, and measure relative L^2 error:

$$\hat{\rho}_i = \bar{\rho} + \sum_{k=1}^K \hat{A}_{ik} \phi_k, \quad e_i = \frac{\|\hat{\rho}_i - \rho_i\|}{\|\rho_i\|}.$$

A surrogate cannot project an unseen field, so LOOCV necessarily tests *POD + regression together*. The total held-out error ($\sim 28\%$) thus has *two* substantial components: a $\sim 10\text{--}14\%$ **basis** error (the linear POD does not fit the curved ensemble well — see the absence of a spectral elbow below) plus the additional regression error on top. Neither component is small; both trace back to the same cause (a curved manifold under-sampled by 20 points). Three lightweight regressors are compared: **rbf** (thin-plate spline), **gp** (small NumPy Gaussian process), **poly** (degree-2).

3 Core results

3.1 Density field

The singular-value spectrum decays *gently, with no elbow at all* (consecutive ratios $\sigma_k/\sigma_1 = 1.00, 0.61, 0.53, 0.43, 0.41, 0.33, \dots$ — a smooth taper, not a knee): 90% energy needs ~ 10 modes and the coverage curve keeps climbing with diminishing returns. This is itself a negative result for linear POD: the field ensemble is *not* low-rank-linear, so there is no natural truncation K and even the best K leaves a non-trivial basis error (Sec. 2.5). This is the expected fingerprint of a transport-dominated / high-Kolmogorov-width problem (density blobs shifted by v_{adv} are nearly L^2 -orthogonal). Normalization barely moves the held-out error. LOOCV is $\sim 28\%$ (rbf); the error concentrates in the high- $|\omega_{\text{pin}}|$ /low- v_{adv} corner and at the sampling convex-hull edges. Restricting to the near-pin first 20% of x (dropping the v_{adv} -sensitive wake) improves the surrogate.

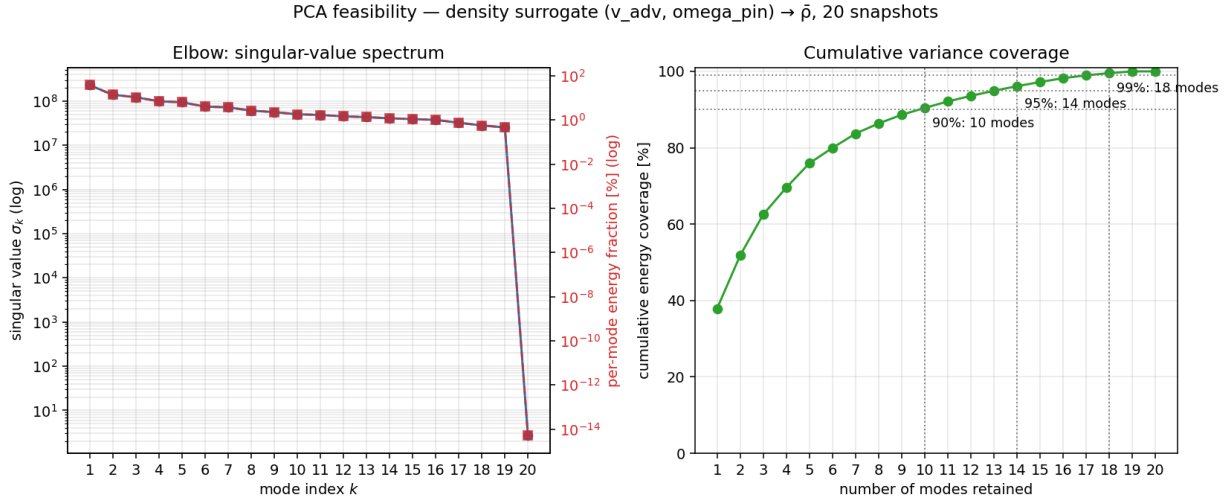


Figure 1: Density POD spectrum. Left: singular-value elbow (mode 17 is the null mode from mean-centring). Right: cumulative energy coverage, with 90/95/99% thresholds annotated.

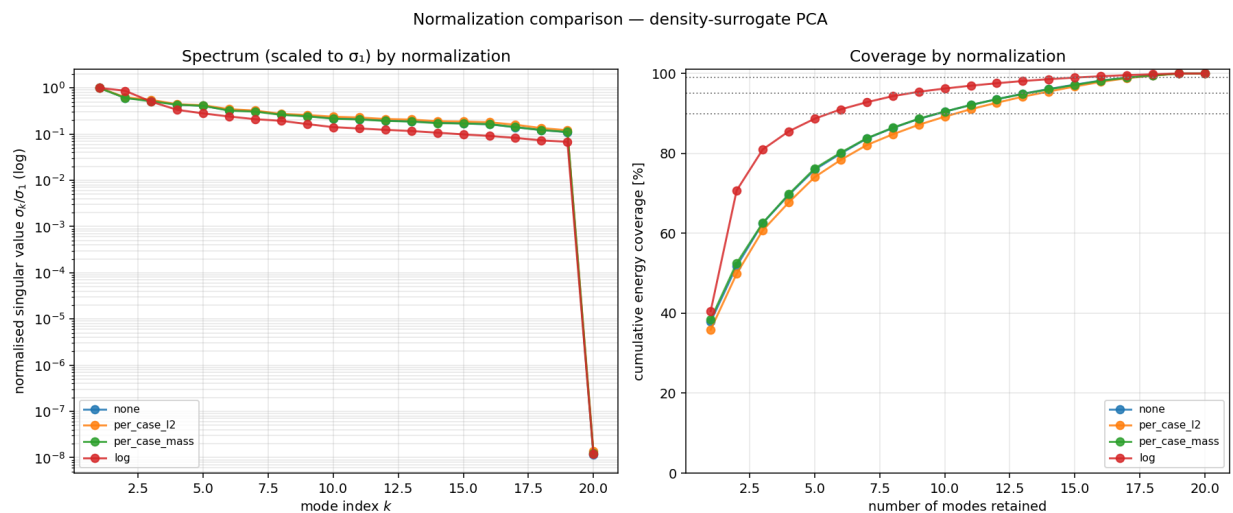


Figure 2: Normalization comparison. `per_case` options barely change the spectrum; `log` reaches coverage in fewer modes but its error is measured in log-density space and is not comparable.

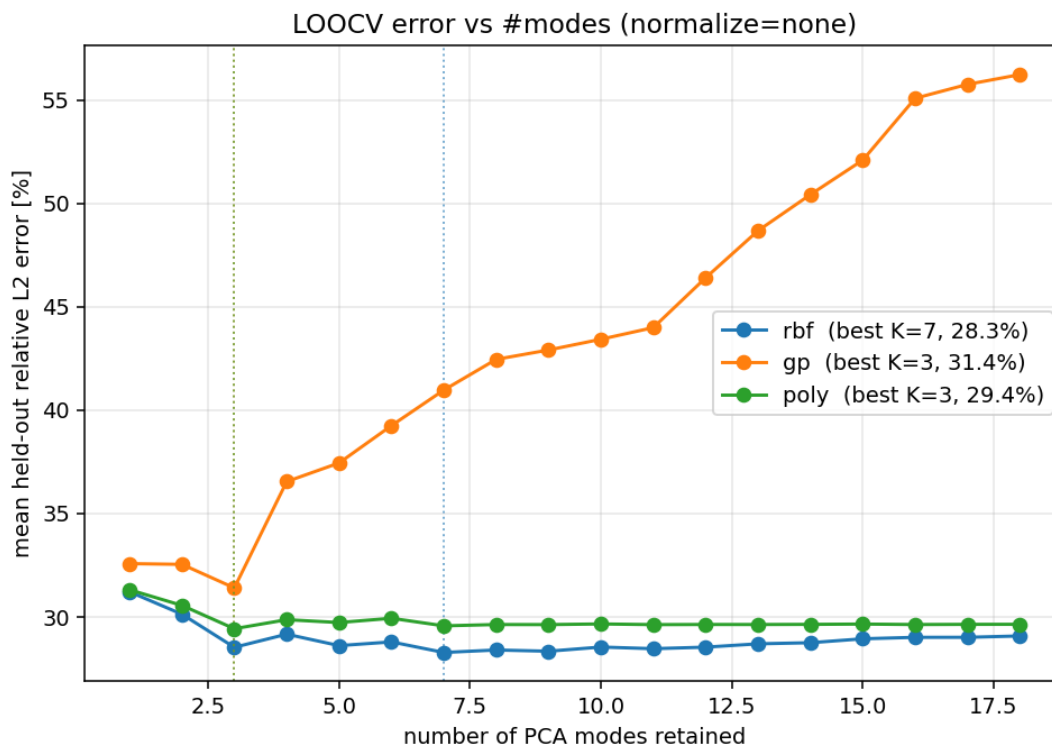


Figure 3: Density LOOCV error vs. number of retained modes, per regressor. The dip (bias/variance sweet spot) is at $K \approx 3-7$; all reasonable regressors converge to a similar floor set by the sample count.

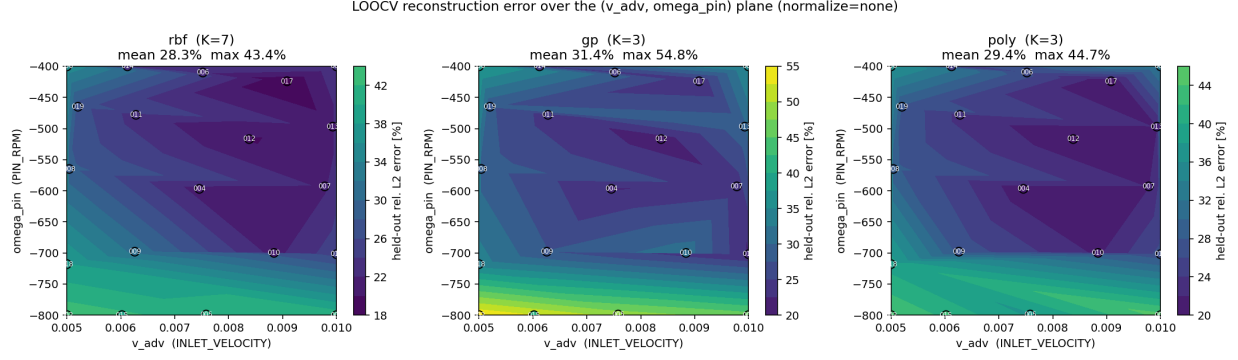


Figure 4: Density held-out error over the (v_{adv}, ω_{pin}) plane. Error peaks in the high- $|\omega_{pin}|$ /low- v_{adv} corner — the sparsely sampled region.

3.2 Particle cloud

Seeds are byte-identical across cases, so the natural target is the per-particle final-step displacement. A subtle finding: because correspondence is exact, subtracting the (constant) bulk drift is a rigid offset fully absorbed by mean-centring, so raw / co-moving / final representations give *identical* POD modes. Thresholding to “eliminate unaffected particles” is moot: $> 97\%$ of particles exceed even $|\xi| > 5\Delta p$, since the seed spacing is tiny relative to typical displacement.

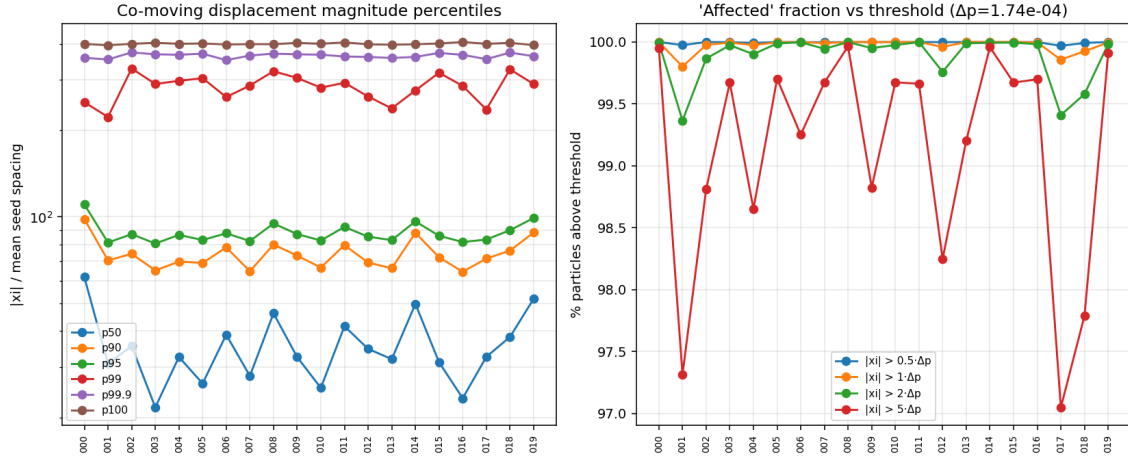


Figure 5: Co-moving displacement magnitude $|\xi|/\Delta p$ per case (left) and the “affected” fraction vs. threshold (right). Almost all particles are “affected”, so a magnitude threshold cannot isolate the stir zone.

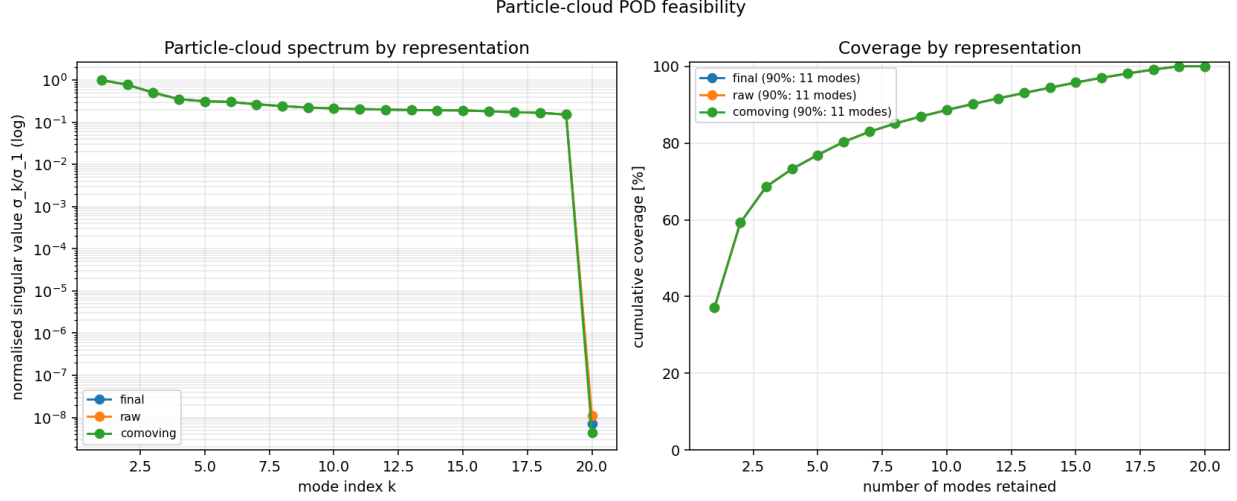


Figure 6: Particle-cloud POD by representation. The **final**, **raw**, and **comoving** curves coincide exactly — drift subtraction does not change the modes.

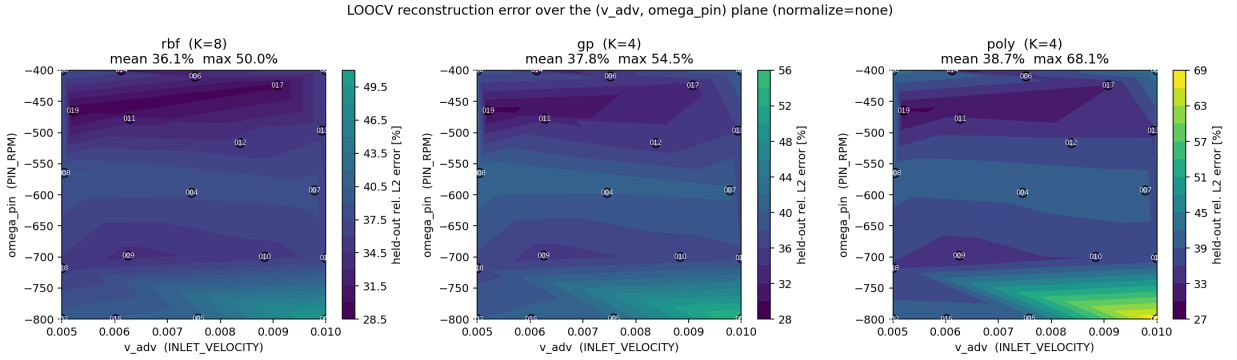


Figure 7: Particle displacement LOOCV error over the (v_{adv}, ω_{pin}) plane.

The Lagrangian displacement is harder to predict than the Eulerian density ($\sim 36\%$ vs. $\sim 28\%$ LOOCV), consistent with the literature: a density field spatially averages many trajectories and is a lower-variance, more predictable quantity.

4 Why the error is what it is: three diagnostics

4.1 Input features and active subspace

Mapping the inputs to physics-motivated features — weld pitch $p_w = v_{adv}/|\omega_{pin}|$, log inputs, and the first-stage ROM (velocity / temperature / pressure) coefficients — *none* beat the raw two-scalar baseline. An active-subspace analysis is consistent with this: the estimated input→coefficient map is *apparently* low-rank, with its dominant direction close to *pure* v_{adv} rather than weld pitch — so if a reduction to one input combination were warranted it would be v_{adv} , not p_w . We caution, however, that at $n=20$ over a 2D input space an “apparently 1D” active subspace cannot be cleanly distinguished from simply having too few samples to resolve the second direction’s effect; small-sample regressions generically look artificially low-rank. The first-stage coefficients are largely a

reparametrisation of the two scalars (leading modes have $R^2 \approx 1.0$ with the inputs; only mode ≥ 3 carries new information, which does not drive the density modes), and adding all of them overfits.

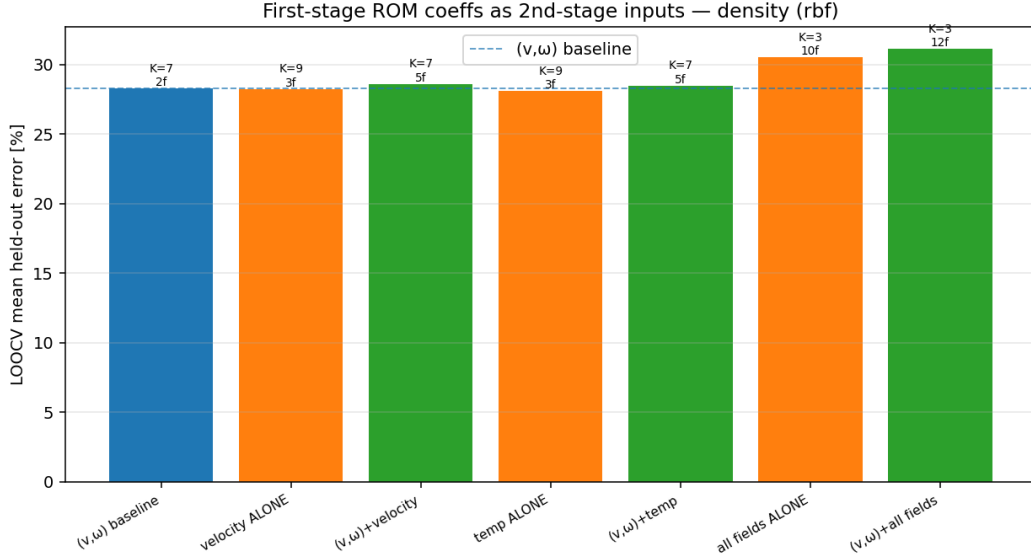


Figure 8: First-stage ROM coefficients as second-stage inputs. Velocity coefficients alone are marginal; piling on all fields overfits (12 features, 20 points $\Rightarrow$ best- K collapses to 3).

4.2 Manifold geometry

Characterising the case manifold with linear/metric methods (before any nonlinear embedding) reveals a *high linear dimension but low intrinsic dimension* — the signature of a smooth but *curved* manifold. Classical MDS axes are the variance-ordered principal directions of the case cloud; we measure their alignment with the inputs afterwards.

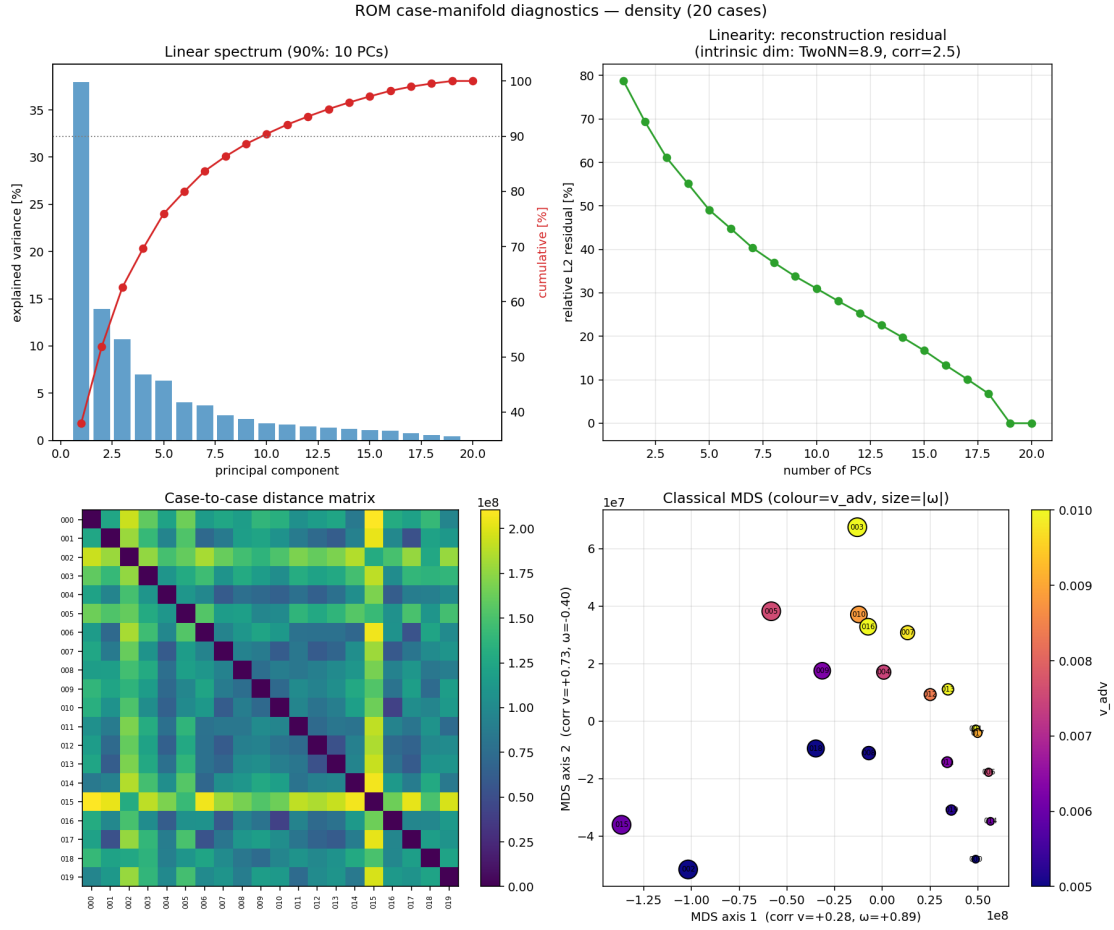


Figure 9: Density case-manifold diagnostics: spectrum, linearity residual, distance matrix, and classical-MDS layout coloured by v_{adv} / sized by $|\omega_{pin}|$. Intrinsic dim ≈ 2.5 vs. 10 linear PCs for 90%.

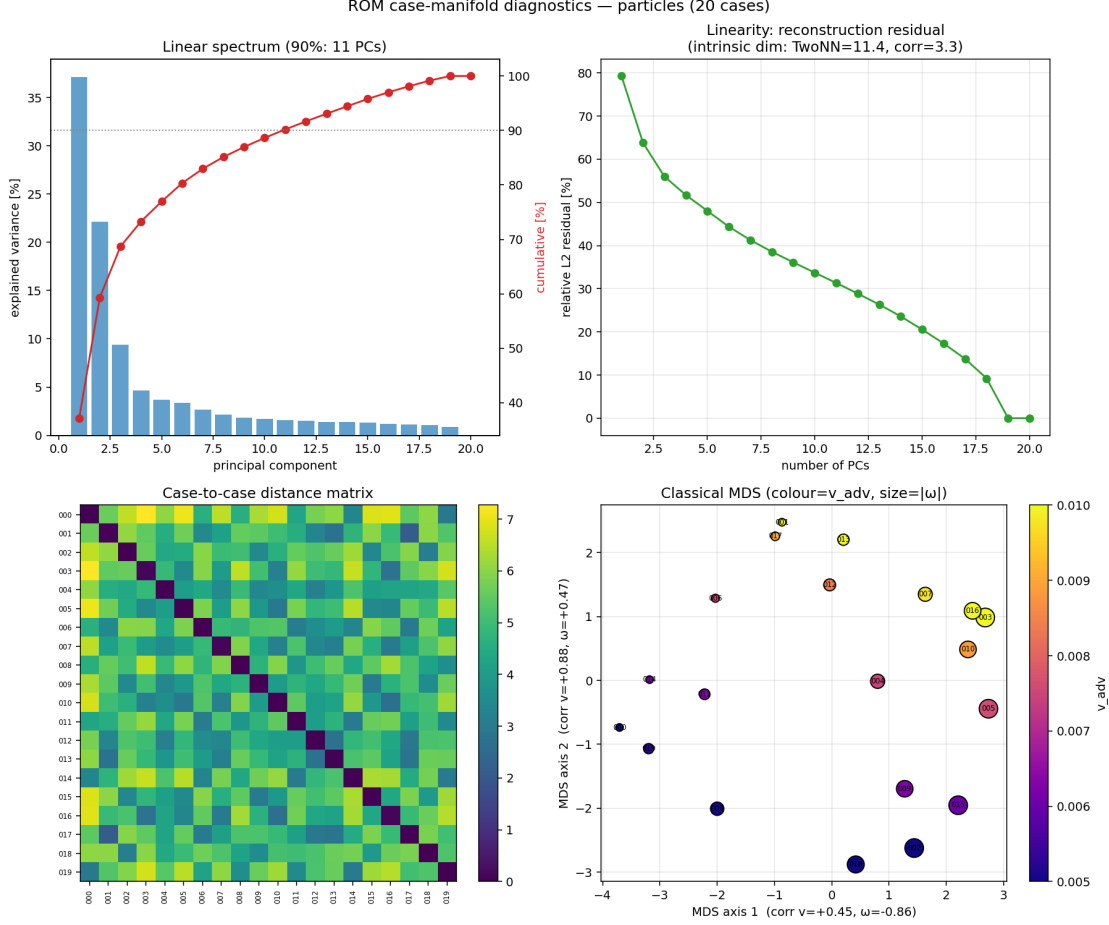


Figure 10: Particle case-manifold diagnostics. MDS axis 1 tracks ω_{pin} strongly; particles give a clean (ω_{pin}, v_{adv}) split, whereas density’s axis 2 mixes the parameters (the wake couples them).

How the MDS axes arise, and why they differ. Classical MDS double-centres the distance matrix, $B = -\frac{1}{2}JD^2J$, and takes coordinates $x_i = \sqrt{\lambda_i}v_i$ for the two largest eigenvalues — so the axes are variance-ordered and data-driven (for Euclidean D this is PCA of the cases), *not* parameter-picked. Both targets have **axis 1** $\approx \omega_{pin}$ (the largest source of case-to-case variation). Particles split cleanly (axis 2 $\approx$ pure v_{adv}); density’s axis 2 *mixes* v_{adv} and ω_{pin} because the field couples them through the wake (length $\propto v_{adv}$, pattern $\propto \omega_{pin}$). A small negative MDS eigenvalue is the expected mild-curvature signal.

4.3 Discrete curvature: how curved, and where

The intrinsic-dimension gap above says the manifold *is* curved; to see *how much* and *where*, we compute discrete secant-vector curvature directly in the high-dimensional snapshot space (not the MDS embedding). Three estimators are used: the **turning angle** between consecutive secants along 1D parameter paths (fix one input, sweep the other); the **Menger curvature** $\kappa = 4 \text{Area}/(abc) = 1/R_{\text{circ}}$ of consecutive triples, scale-normalised by the median pairwise distance so it is dimensionless and comparable across manifolds; and a **local-PCA** curvature (trailing k -NN eigenvalue energy beyond the intrinsic dimension). The estimators were validated on known geometry (a straight line gives 0; a circle of radius R gives Menger = $1/R$ exactly).

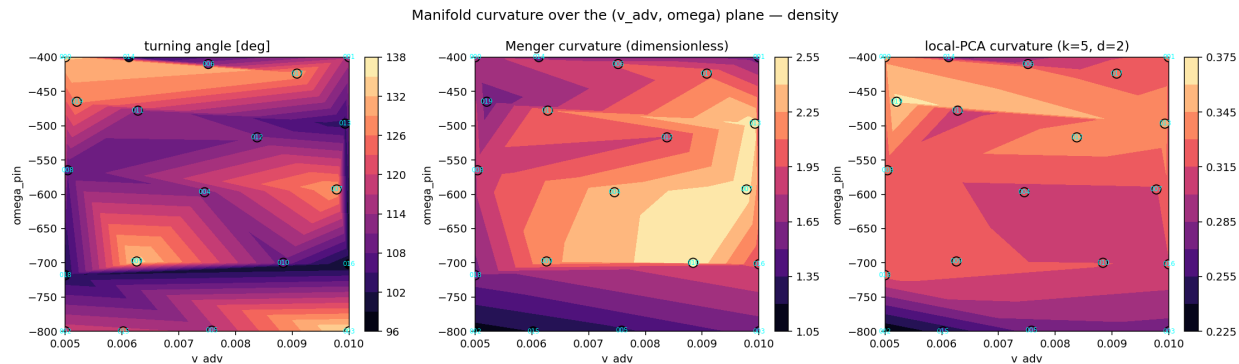


Figure 11: Density manifold curvature over the $(v_{\text{adv}}, \omega_{\text{pin}})$ plane (turning angle, Menger, local-PCA). Curvature is *highest in the interior* (mid/high v_{adv}) and *lowest along the $\omega_{\text{pin}} = -800$ sampling edge*.

One observation is robust; a second is only suggestive.

- **(Robust) The manifold is smooth, not sharply bent.** Turning angles sit at $\sim 117^\circ$ for every target — the honest high-dimensional signature that consecutive secants are near-orthogonal; there are no sharp folds. Scale-normalised Menger is ~ 1.7 – 1.9 across all four representations (density/particle, 3D/2D), i.e. comparable and moderate. This says the response varies smoothly with the inputs — good for interpolation, and an argument *against* the presence of discontinuities that only a nonlinear method could handle.
- **(Suggestive) Error appears to concentrate on the sampling boundary.** Per-case LOOCV error correlates *negatively* with Menger (-0.74) and local-PCA (-0.58) curvature: the highest-error cases (002, 003, 005, 015, all at $\omega_{\text{pin}} \approx -800$) sit on the convex-hull boundary and have *low* curvature, while the lowest-error cases are interior and most curved (Fig. 11). We deliberately do *not* treat this as an established mechanism: both the curvature correlation and the per-case LOOCV error are computed from ~ 20 points, and LOOCV error is itself a high-variance, single-held-out estimate (here the per-case error spans 18–43%). A correlation of -0.74 on 20 noisy points has wide sampling uncertainty, and the pattern is equally consistent with the simpler “too few samples, period.” The local-PCA estimator’s cross-manifold ranking is unreliable at $n=20$ (used only for the within-manifold pattern).

Interpretation (a hypothesis, not a diagnosis). The most *parsimonious* explanation of the projection-vs-LOOCV gap is degrees-of-freedom starvation: retaining ~ 7 – 10 modes leaves only ~ 10 – 13 effective DOF for the regression, fitted from 20 points — a very thin margin that alone predicts large held-out error. On top of that, the geometry *hints* that the residual error is worst where the model must extrapolate past the sampling boundary rather than in high-curvature interior regions. Both readings point the same practical way (more, better-placed samples), so we adopt the recommendation without committing to the curvature-based causal story as fact.

4.4 The bottleneck

The total $\sim 28\%$ error is limited on *two* fronts. First, the linear POD **basis** itself is inadequate: the spectrum has no elbow and the projection (own-coefficient) error is ~ 10 – 14% at usable K , because the transport-dominated ensemble is not low-rank in a linear basis. Second, the **regression** adds error on top, most simply from **degrees-of-freedom starvation**: 20 samples cannot robustly fit ~ 7 – 10 mode coefficients plus the interpolation between them. Consistent with this, no lever that

does *not* add information helps — normalization, input features, and regressor choice all converge to a similar floor, and every LOOCV number carries a standard error of order a few percentage points (e.g. density 28.3% has per-case spread 18–43%), so the point estimates in the tables should be read as $\pm$ several points, not exact. The suggestive boundary/curvature pattern (Sec. 4.3) refines *where* to add samples but is not needed for the core conclusion: both the basis inadequacy and the regression starvation stem from **too few samples of a curved response**, and both improve only with more data.

5 Data-quality fixes (all 20 cases)

Three cases were initially set aside and later fixed, after which the entire study runs on all 20. Cases 000/001 produced ~ 1.4 – 2.5% *runaway particles* (flying to $x \approx 2$ – 3.7 m vs. a ~ 0.08 m domain); the density driver sized its voxel grid from the trajectory bbox, which those runaways inflated to 10^{11} – 10^{12} voxels $\Rightarrow$ GPU out-of-memory. A clean rerun (plus an absolute physical ROI box) removed them. Case 002’s density had been *host*-RAM OOM-killed (batch dedup over 8001 steps $\times$ 360k particles ≈ 2.9 billion points > 58 GiB); recomputing with a step stride fixed it. Including the (now clean) cases barely moved any number, confirming the earlier exclusions did not bias the study.

6 Representation studies

6.1 Final-step vs. union (time-averaged) density

The core density study uses the *union* density (time-average over the whole trajectory). Computing instead a *final-step* density (one snapshot) and re-running the suite shows the union is markedly better for ROM (Table 2). Temporal averaging denoises the case-to-case stochastic variability of any single instant, so the union field varies more smoothly and lower-dimensionally with the inputs.

metric	union (time-avg)	final-step
PCA modes for 90%	10	12
PC-1 energy	37.9%	25.5%
LOOCV, full domain (rbf)	28.3%	49.3%
LOOCV, near-pin (rbf)	24.6%	75.2%
manifold intrinsic dim	2.5	4.8

Table 2: Union density is substantially more predictable than a single final snapshot.

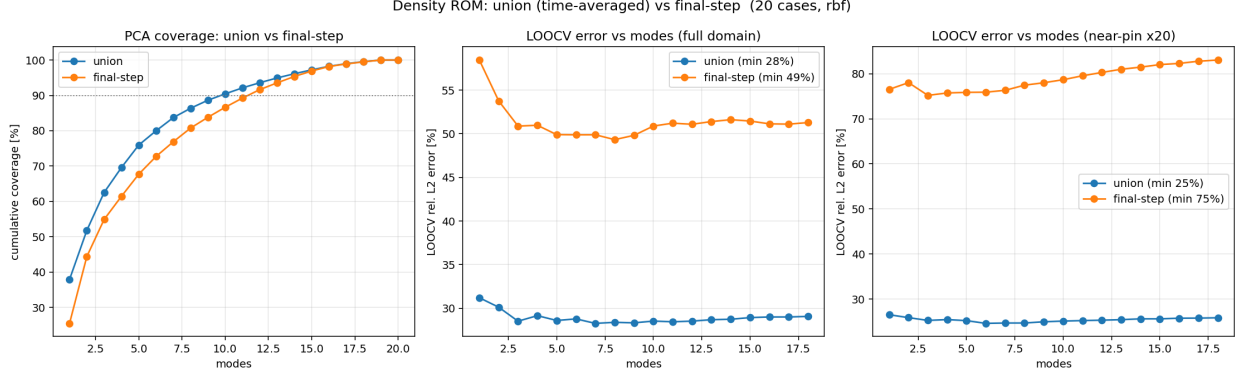


Figure 12: Union vs. final-step density: coverage curves (left) and LOOCV error vs. modes for the full domain (centre) and near-pin (right).

6.2 2D (y, z) cross-section (collapse the advection axis)

The wake stretches along x and its *length* scales with v_{adv} — the very feature that dominates the error. Collapsing x (summing the slab into a (y, z) cross-section, a $\sim 300\times$ reduction) tests whether integrating out that axis helps. It **helps density and hurts particles**, for the same underlying reason (Table 3).

	3D	2D (y, z)
density — PCA modes for 90%	10	6
density — LOOCV (rbf)	28.3%	20.6%
density — intrinsic dim	2.5	1.7
particles — PCA modes for 90%	11	14
particles — LOOCV (rbf)	36.1%	41.4%
particles — intrinsic dim	3.3	4.3

Table 3: 2D cross-section: the best density result of the study, but worse for particles.

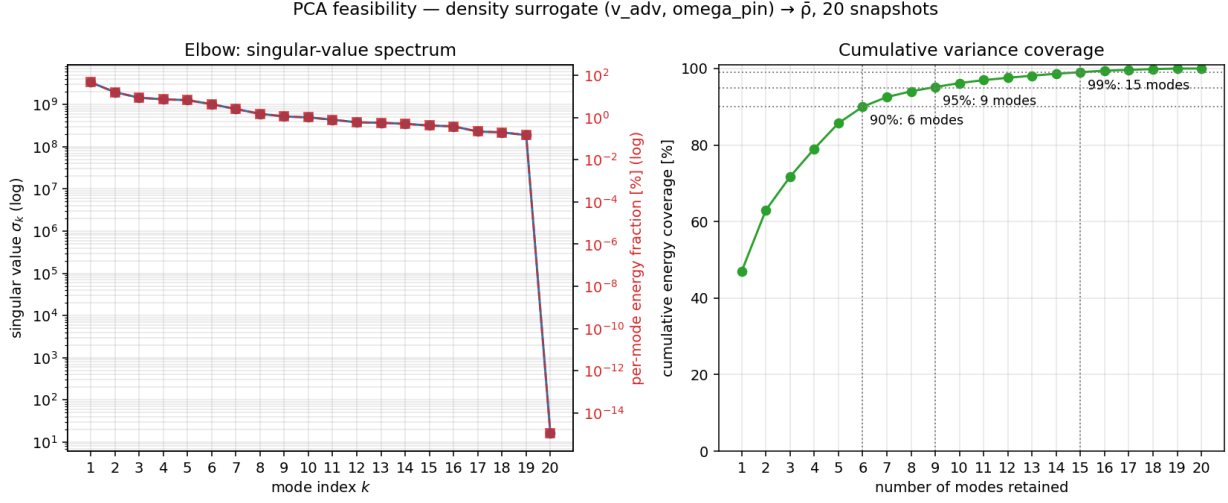


Figure 13: 2D density POD spectrum — markedly faster decay than 3D (90% in 6 modes).

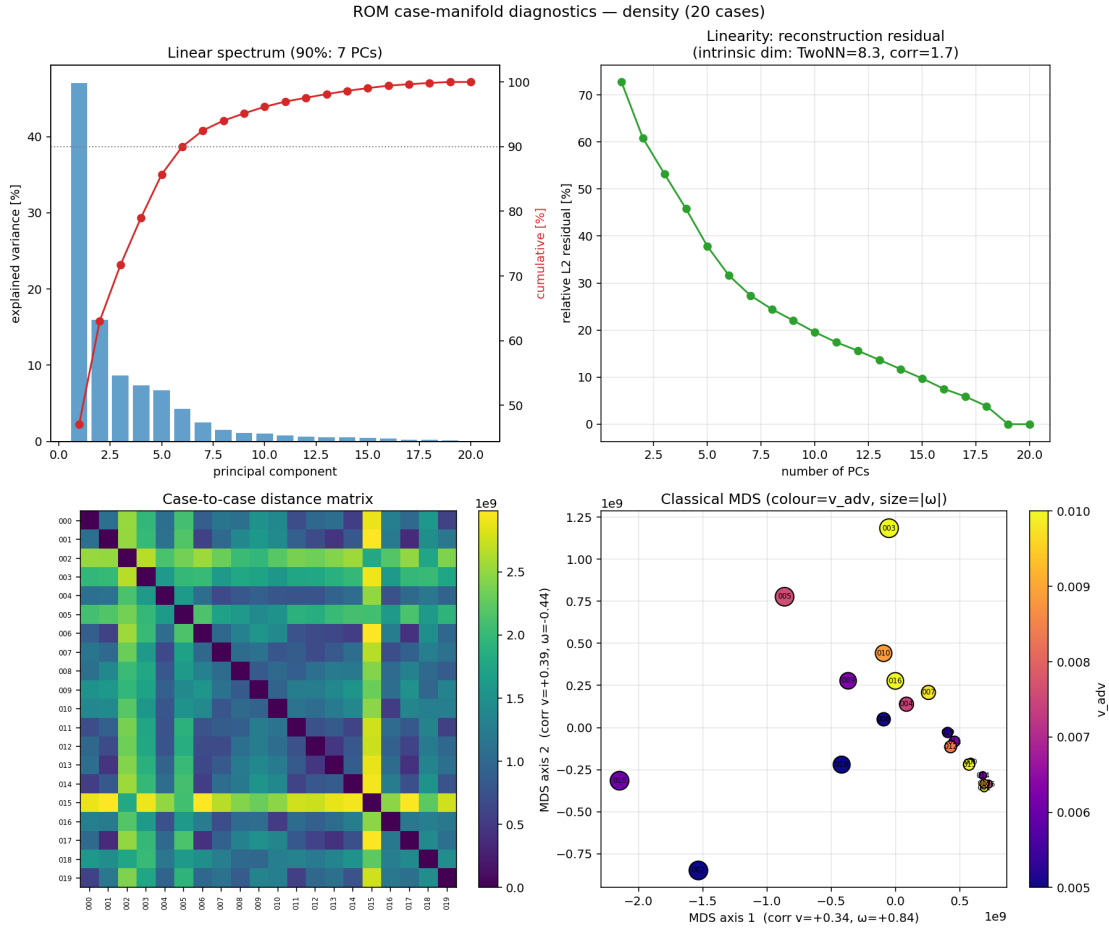


Figure 14: 2D density case-manifold: intrinsic dim ≈ 1.7 , a cleaner and lower-dimensional manifold than 3D.

For **density**, integrating along x averages out the v_{adv} -driven wake variability and per-voxel

noise (the spatial analogue of the union’s temporal denoising). For **particles**, x is the large, smooth, easily-predicted advection displacement; dropping it removes the “free” mode and leaves only the small, chaotic in-plane stir motion, which is harder to predict. So for the cloud, x is signal, not noise.

Caveat on the 2D “improvement.” Summing out x is not a free lunch: it reduces the predicted field’s information content by $\sim 300\times$. Part of the lower LOOCV error therefore reflects predicting an *easier, less-informative target* (the cross-section integrates away exactly the v_{adv} -dependent wake the user may care about), not only a genuinely better ROM. The 2D result is the right choice *if* the cross-section is the quantity of interest; it is not evidence that the surrogate “understands” the full field better.

7 Summary of variants

target	variant	90% modes	best LOOCV	note
density	union, full domain	10	28.3%	baseline (all 20)
density	union, near-pin x	11	24.4%	drops the v_{adv} -wake
density	final-step	12	49.3%	time-avg far better
density	2D (y, z)	6	20.6%	lowest error, but easier target
particles	3D comoving	11	36.1%	harder than density
particles	2D (y, z)	14	41.4%	worse (x is signal)
<i>negative results (do not help):</i> weld-pitch inputs, log inputs, first-stage ROM coeffs, richer regressors.				

Table 4: All main variants (rbf regressor, 20 cases). LOOCV = mean held-out relative L^2 error.

8 Conclusions and next steps

1. **Linear POD is a workable first approximation, not a validated model of a linear response.** It gives ~ 21 – 28% held-out density error (per-case spread wide, e.g. 18–43%) — useful as a baseline, but the underlying manifold is *curved* (intrinsic dimension ~ 2.5 vs. ~ 10 linear modes for 90% energy). Linear POD works here because it is a convenient global basis, not because the response is linear; that mismatch is a real, not merely cosmetic, limitation.
2. **Both the basis and the regression are limited — and both point to more data.** The $\sim 28\%$ held-out error is not purely a regression failure: the linear POD *basis* itself carries a ~ 10 – 14% error with no spectral elbow, because the curved ensemble is not low-rank in a linear basis. The additional regression error is most simply explained by degrees-of-freedom starvation (~ 7 – 10 coefficients from 20 points). No information-free lever (normalization, input features, regressor) helps, and all LOOCV numbers carry a standard error of several points. Whether the residual error is *specifically* a boundary/extrapolation effect (Sec. 4.3) is a plausible but *unconfirmed* hypothesis at $n=20$, not an established mechanism.
3. **Representation matters (within limits).** The *time-averaged* density is more predictable than a final snapshot, and the *2D* (y, z) *cross-section* reaches $\sim 21\%$ — but the latter also predicts a $\sim 300\times$ smaller, lower-information target, so it is a good choice only if the cross-section is the quantity of interest.

4. **More data is the prerequisite for everything else.** The manifold is curved but smooth (no sharp folds), which is encouraging — but 20 points is too few to (i) fit the linear regression well, (ii) confidently diagnose *where* and *why* the error concentrates, or (iii) support any nonlinear alternative. Methods such as *local / patch POD*, *manifold learning* (t-SNE, UMAP), and *autoencoders* would each overfit at $n=20$; they only become viable, and their value only becomes testable, with substantially more samples (the literature suggests ~ 50 – 100 for a 2D parameter space of this complexity).

Recommended next step. Acquire more samples, and place them to serve two goals at once: (a) *extend and densify the sampling envelope at its edges* (particularly the $\omega_{\text{pin}} \approx -800$ /low- v_{adv} boundary, where the largest LOOCV errors sit), converting extrapolation into interpolation; and (b) *add samples where the curvature changes*, so the manifold’s shape — and hence the choice between linear POD, patch POD, and nonlinear reduction — can actually be resolved rather than guessed. One suggestive (not conclusive) data point: adding the previously-missing boundary case reduced the worst-case density error from $\sim 48\%$ to $\sim 43\%$; a larger, deliberately-placed batch is needed to confirm the strategy and to unlock the nonlinear methods above.

Reproducibility

All code is in the `jaxtrace/rom/` package (branch `feature/rom-svd`), with drivers `run_rom_pca.py`, `run_rom_loocv.py`, `run_rom_particles.py`, `run_rom_first_stage.py`, `run_rom_manifold.py`, `run_rom_curvature.py`, and `run_rom_density_compare.py`. Density product and 2D projection are selected via `--density-filename` and `--project-2d sum`; particle 2D via `--components y z`. The companion Markdown report `docs/rom_feasibility_report.md` contains the full step-by-step derivations and the complete figure set.